

UNIVERSITY OF AZAD JAMMU & KASHMIR CHELLAH
CAMPUS MUZAFFERABAD



DEPARTMENT OF SOFTWARE ENGINEERING

Databases Systems

(Lab Task # 02)

Submitted By:

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(2024-SE-37)

Submitted To:

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Session: 2024-2028

Implementation of a Point of Sale (POS) Database System

➤ Introduction:

- The objective of this lab is to design a robust relational database for a Point of Sale (POS) system.
- The system manages products, stock levels, and customer transactions across multiple categories with defined user roles.

➤ Database Components:

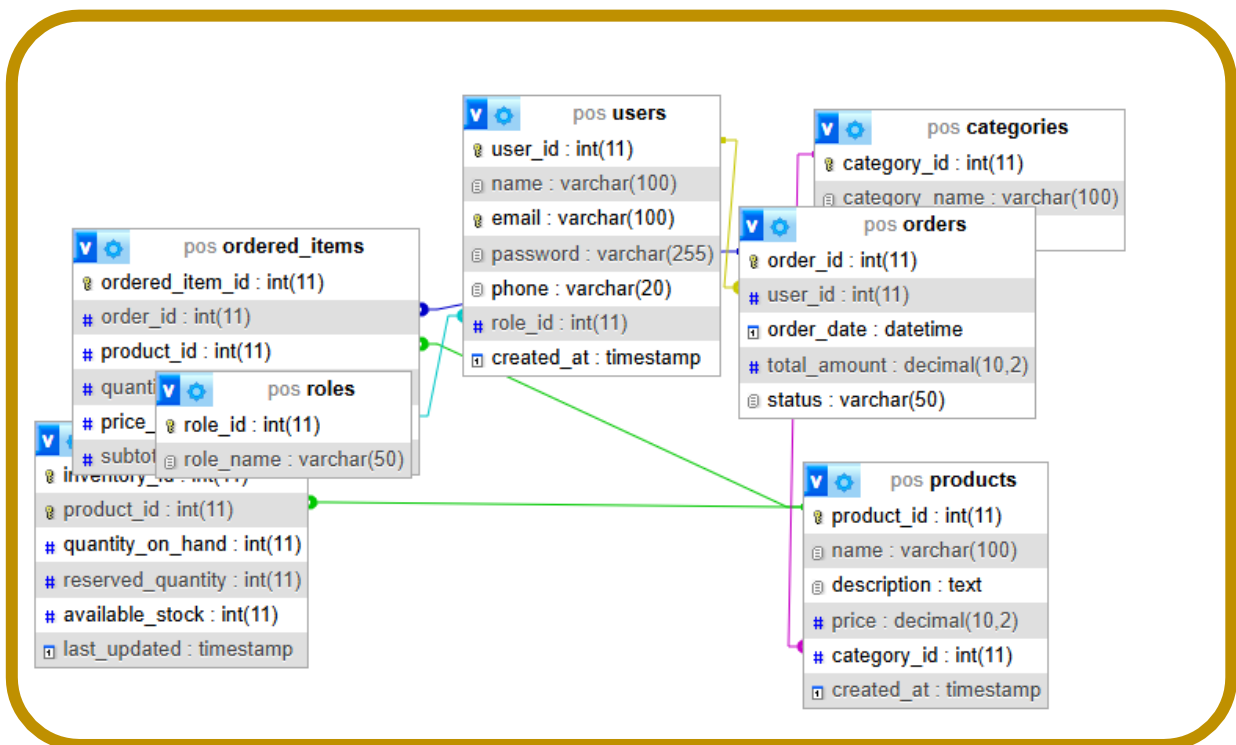
- **User Roles:** The system includes Admin, Salesman, and Customer roles with specific access levels.
- **Product Catalog:** Items like Electronics, Clothing, and Groceries are organized into dedicated categories.
- **Inventory Tracking:** Uses a smart calculation system to track 'Available Stock' automatically.
- **Sales Records:** Tracks every order with a timestamp, total amount, and itemized billing.

➤ Technical Features:

- **Primary & Foreign Keys:** Used to link products to categories and orders to customers.
- **Data Integrity:** Implemented 'Cascading' so that updates to products automatically reflect in the inventory.
- **Financial Precision:** All prices and totals use decimal formatting for 100% accuracy.

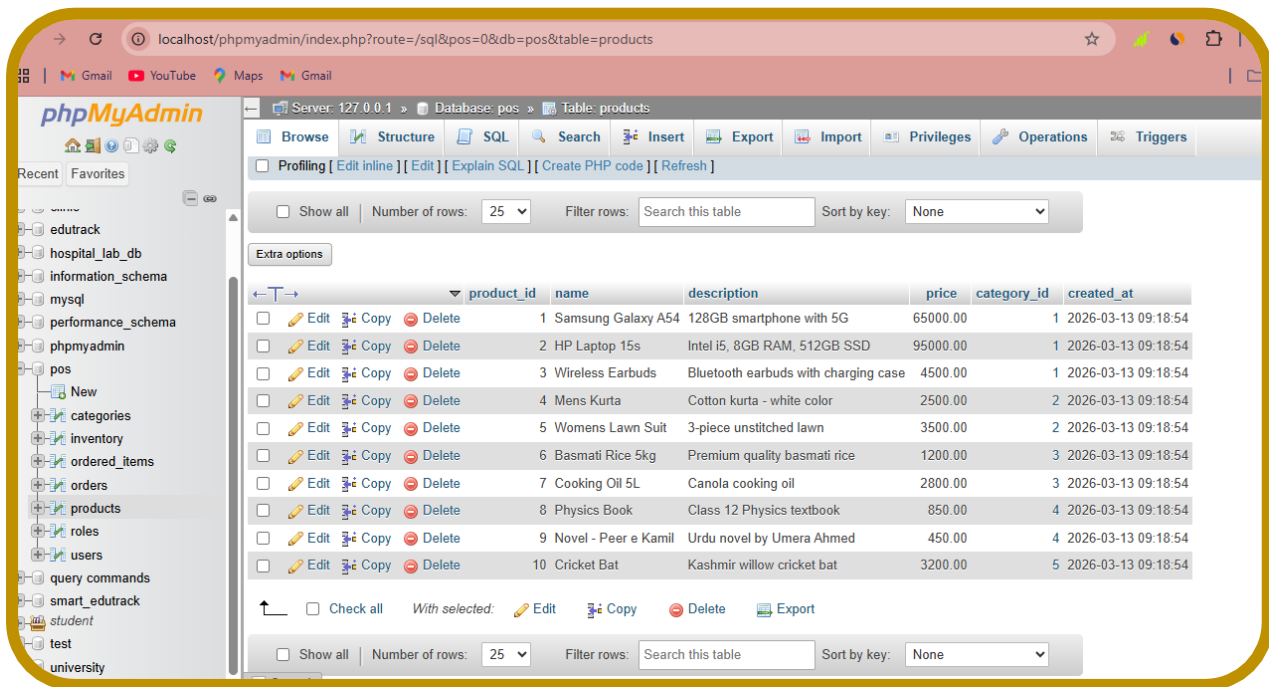
➤ Implementation Screenshots:

1: Database Relationships



- **Description:** This screenshot shows how all 7 tables are connected in the database.

2:- Table Records (Products & Users)

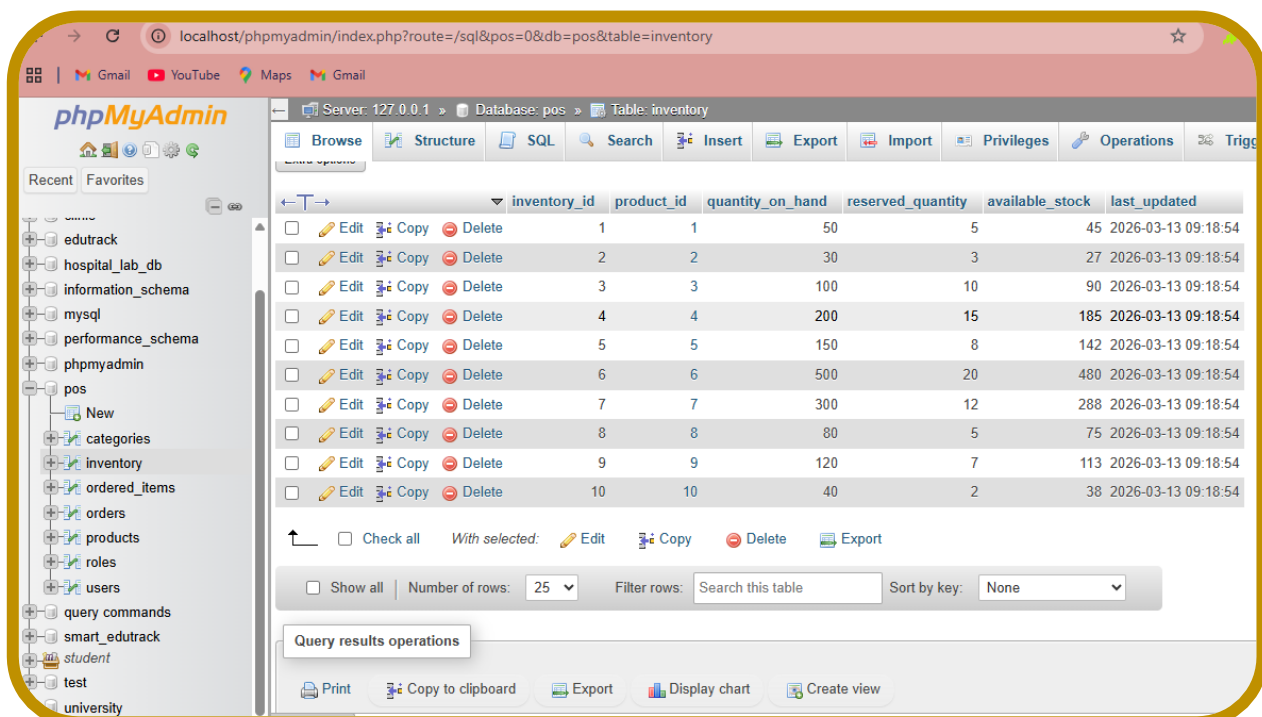


The screenshot shows the phpMyAdmin interface for the 'pos' database, specifically the 'products' table. The table contains 10 records, each with a product ID, name, description, price, category ID, and creation timestamp. The interface includes a sidebar with a database structure tree, a top navigation bar with various tools (Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, Operations, Triggers), and a main content area with table controls and a data grid.

product_id	name	description	price	category_id	created_at
1	Samsung Galaxy A54	128GB smartphone with 5G	65000.00	1	2026-03-13 09:18:54
2	HP Laptop 15s	Intel i5, 8GB RAM, 512GB SSD	95000.00	1	2026-03-13 09:18:54
3	Wireless Earbuds	Bluetooth earbuds with charging case	4500.00	1	2026-03-13 09:18:54
4	Mens Kurta	Cotton kurta - white color	2500.00	2	2026-03-13 09:18:54
5	Womens Lawn Suit	3-piece unstitched lawn	3500.00	2	2026-03-13 09:18:54
6	Basmati Rice 5kg	Premium quality basmati rice	1200.00	3	2026-03-13 09:18:54
7	Cooking Oil 5L	Canola cooking oil	2800.00	3	2026-03-13 09:18:54
8	Physics Book	Class 12 Physics textbook	850.00	4	2026-03-13 09:18:54
9	Novel - Peer e Kamil	Urdu novel by Umera Ahmed	450.00	4	2026-03-13 09:18:54
10	Cricket Bat	Kashmir willow cricket bat	3200.00	5	2026-03-13 09:18:54

- **Description:** Verification of 10+ records successfully inserted into the system.

3:- Inventory Management



The screenshot shows the phpMyAdmin interface for the 'pos' database, specifically the 'inventory' table. The table contains 10 records, each with an inventory ID, product ID, quantity on hand, reserved quantity, available stock, and last updated timestamp. The interface includes a sidebar with a database structure tree, a top navigation bar with various tools (Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, Operations, Triggers), and a main content area with table controls and a data grid.

inventory_id	product_id	quantity_on_hand	reserved_quantity	available_stock	last_updated
1	1	50	5	45	2026-03-13 09:18:54
2	2	30	3	27	2026-03-13 09:18:54
3	3	100	10	90	2026-03-13 09:18:54
4	4	200	15	185	2026-03-13 09:18:54
5	5	150	8	142	2026-03-13 09:18:54
6	6	500	20	480	2026-03-13 09:18:54
7	7	300	12	288	2026-03-13 09:18:54
8	8	80	5	75	2026-03-13 09:18:54
9	9	120	7	113	2026-03-13 09:18:54
10	10	40	2	38	2026-03-13 09:18:54

- **Description:** Proof of the automated stock calculation functioning correctly.